

The DOM

Meaning: *The browser turns your webpage into a JavaScript object. This lets you use JS to control the page.*

1. Selecting Elements (querySelector)

Meaning: *Searches the page and grabs an HTML element so you can use it in JS.*

```
document.querySelector("p")
```

Grabs the **first** `<p>` tag.

```
document.querySelector(".box")
```

Grabs the first element with **class="box"**.

```
document.querySelector("#id")
```

Grabs the exact element with **id="id"**.

2. Changing Text & HTML

```
element.innerHTML
```

Reads or changes content, **including HTML tags**.

```
element.innerText
```

Reads or changes **only visible text**. Ignores HTML tags.

Input Boxes (.value)

Meaning: `<input>` boxes don't have "inside" content. Text is stored in **.value**.

```
document.querySelector("input").value
```

Read what the user typed.

```
document.querySelector("input").value = "hi"
```

Write "hi" into the textbox.

⚠ Never use `innerHTML` for inputs! Use `.value`

3. Values & Coercion

Meaning: *Everything from the DOM is a **String**. JS secretly converts Strings to Numbers to help you do math (Coercion).*

```
Number(value)
```

Force a string to become a real Number.

```
'25' - 5
```

→ **20** (JS auto-converts for `-`, `*`, `/`)

```
'25' + 5
```

→ **"255"** ⚠ `+` joins strings! Use `Number()` first.

4. Events & Event Object

Events: *User actions (like clicking).* **Listeners:** *JS waiting for it to happen.*

Common Listeners:

`onclick`, `onkeydown`, `onscroll`, `onmouseenter`

Event Object: *An automatic package of info telling you exactly what happened.*

```
element.onclick = function(event) {  
  console.log(event.key);  
  console.log(event.target);  
}
```

event is the package containing the details.

Which keyboard key was pressed.

The exact element that triggered the event.